

Lesson 8. Partial Differential Equations I

University of Budapest

Faculty of Informatics, Department of Numerical Analysis

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A **partial differential equation (PDE)** is an equation containing partial derivatives of the dependent variable.

In general, we may write a PDE as

$$F\left(u, u_{x_1}, \dots, u_{x_n}, u_{x_1 x_1}, \dots, u_{x_{i_1} \dots x_{i_N}}, x_1, x_2, \dots, x_n\right) = 0 \quad (1)$$

$$i_1, \dots, i_N \in \{1, 2, \dots, n\}$$

for some function F .

Here x_i are the independent variables and u is the unknown function of these variables.

For example, the following are PDEs

$$u_t + cu_x = 0 \quad (1.1)$$

$$u_{xx} + u_{yy} = x^2 - y^2 \quad (1.2)$$

$$\alpha u_{xx} + 2u_{xy} + 3x^2 u_{yy} = 4e^x \quad (1.3)$$

$$u_x u_{xx} + (u_y)^2 = 0 \quad (1.4)$$

$$(u_{xx})^2 + u_{yy} + \alpha u_x + \beta u = 0 \quad (1.5)$$

We use subscript to mean differentiation with respect to the variables given.

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We use subscript to mean differentiation with respect to the variables given.

The **order** of a PDE is the order of the highest order derivative in the equation.

For example, (1.1) is of first order and (1.2) - (1.5) are of second order.

A PDE is **linear** if it is linear in the unknown function and all its derivatives with coefficients depending only on the independent variables.

For example, (1.1) - (1.3) are linear PDEs.

A PDE is **nonlinear** if it is not linear.

For example, (1.4) and (1.5) are nonlinear PDEs.

A special class of nonlinear PDEs are called quasilinear.

A PDE is **quasilinear** if it is linear in the highest order derivatives with coefficients depending on the independent variables, the unknown function and its derivatives of order lower than the order of the equation.

For example, (1.4) is a quasilinear second order PDE, but (1.5) is not.

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A PDE is called **homogeneous** if the equation does not contain a term independent of the unknown function and its derivatives. Otherwise, the PDE is called **inhomogeneous**.

For example, (1.1), (1.4) and (1.5) are homogeneous.

A **solution** of a partial differential equation is any function (of several variables) that satisfies the equation identically.

A **general solution** is a solution that contains a number of arbitrary independent functions equal to the order of the equation.

A **particular solution** is one that is obtained from the general solution by a particular choice of arbitrary functions.

A **singular solution** is one that cannot be obtained from the general solution by a particular choice of arbitrary functions.

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Example 1: First Order PDE

Consider the partial differential equation:

$$u_t + cu_x = 0$$

The **general solution** is of the form:

$$u(x, t) = f(x - ct)$$

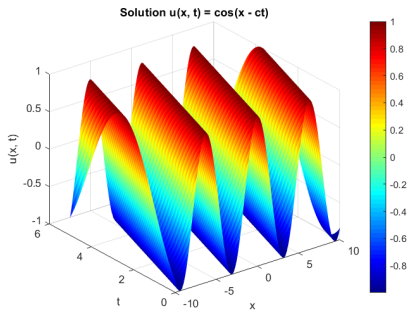
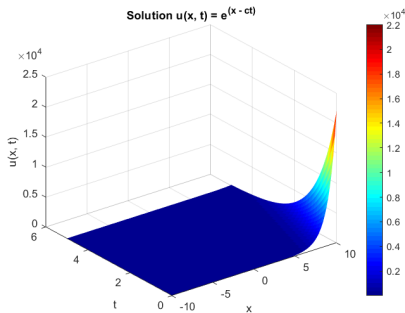
where f is an **arbitrary function**.

A particular solution can be, for example:

$$u(x, t) = e^{x-ct} \quad \text{or} \quad u(x, t) = \cos(x - ct)$$

The fact that these are solutions of the equation can be easily verified.

Example 1: First Order PDE



Example 2: Second Order PDE

Consider the partial differential equation:

$$\frac{\partial^2 u}{\partial x \partial y} = 2x - y$$

The general solution is of the form:

$$u = x^2 y - \frac{1}{2} x y^2 + F(x) + G(y),$$

where $F(x), G(y)$ are arbitrary functions.

With $F(x) = 2 \sin x$ and $G(y) = 3y^4 - 5$, the particular solution is

$$u = x^2 y - \frac{1}{2} x y^2 + 2 \sin x + 3y^4 - 5.$$

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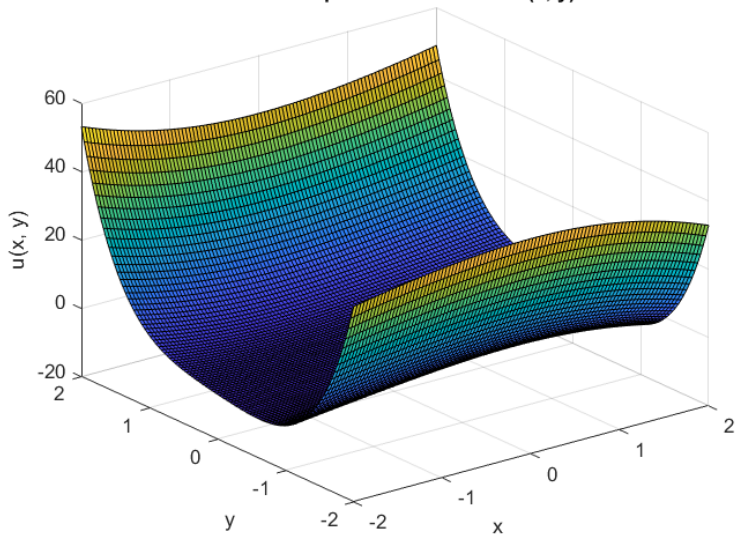
Plot of the particular solution $u(x, y)$ 

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For a function of two variables, a general **first-order partial differential equation** for $u = u(t, x)$ is given as

$$F(t, x, u, u_t, u_x) = 0, \quad (t, x) \in D \subset \mathbb{R}^2 \quad (2)$$

where

- $u_t = \frac{\partial u}{\partial t}$ and $u_x = \frac{\partial u}{\partial x}$ represent the partial derivatives of u with respect to t and x , respectively;
- D is the region in the (t, x) -plane where the independent variables t and x are defined.

The **initial condition** is typically specified at $t = t_0$ for all x in a domain $D \subseteq \mathbb{R}$. It takes the form:

$$u(t_0, x) = g(x), \quad x \in D,$$

where $g(x)$ is a given function that prescribes the initial state of u over the spatial domain D .

The **initial value problem (IVP)** for a partial differential equation (PDE) involves finding a solution $u(t, x)$ that satisfies both the PDE and the initial conditions specified at an initial time.

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A first-order PDE for an unknown function $u(x, y)$ is said to be **linear** if it can be expressed in the form:

$$a(x, y)u_x + b(x, y)u_y + c(x, y)u = f(x, y) \quad (2.1)$$

(We restrict ourselves to unknown functions that depend only on two variables).

Linear PDEs can further be divided into three types:

- **Linear PDE with constant coefficients:**

$$au_x + bu_y = f(x, y)$$

- **Linear PDE with variable coefficients:**

$$a(x, y)u_x + b(x, y)u_y = f(x, y)$$

- **Linear PDE with variable coefficients involving the unknown function explicitly:**

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Linear PDE with Constant Coefficients

$$u_t + cu_x = 0$$

Physical Meaning: This is the **transport** (or **advection**) equation, which models the propagation of a wave or a signal at a constant speed c . It's used to describe phenomena like sound waves moving through a medium at a fixed velocity, or the movement of a pollutant in a river with constant flow velocity.

Linear PDE with Variable Coefficient

$$u_t + a(x)u_x = 0$$

Physical Meaning: This form is often used in **transport equations with spatially varying speed**. For example, $a(x)$ could represent the speed of a wave that changes with position, such as sound propagation in a non-homogeneous medium where the density or temperature varies with location.

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Linear PDE with Variable Coefficients Involving the Unknown Function Explicitly

$$u_t + a(x, t)u_x + b(x, t)u = 0$$

Physical Meaning: This equation can describe **damped** or **amplified waves** where the function $b(x, t)$ represents a damping (or amplification) factor that varies with time and/or space. For example, this could model sound waves attenuating as they pass through different materials, where $b(x, t)$ is a function representing the damping effect in each region.

A first-order PDE for an unknown function $u(x, y)$ is said to be **semi-linear** if it can be expressed in the form:

$$a(x, y)u_x + b(x, y)u_y = f(x, y, u) \quad (2.2)$$

Consider the initial value problem:

$$u_t - 3u^2 = 0, u(0, x) = e^{-x^2}.$$

where $u(0, x) = e^{-x^2}$ is the initial condition

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$$a(x, y, u)u_x + b(x, y, u)u_y = f(x, y, u) \quad (2.3)$$

The inviscid Burgers' equation

The equation describes nonlinear wave propagation, where regions with higher values of u travel faster, leading to wave steepening and shock formation:

$$u_t + uu_x = 0,$$

where

- $u(t, x)$ represents the quantity being transported (such as fluid velocity, car density, or wave height) as a function of time t and position x .
- Nonlinear term uu_x means the speed of transport depends on u itself, causing the wave to steepen and potentially form shocks over time.

A PDE that is not linear, semi-linear, or quasi-linear is said to be **nonlinear**.

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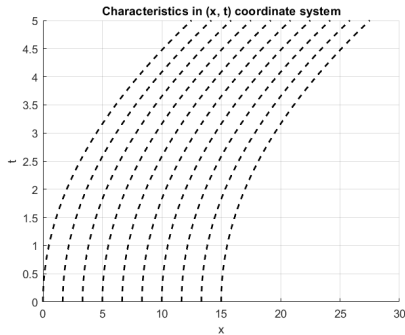
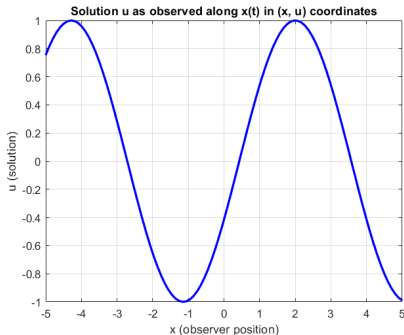
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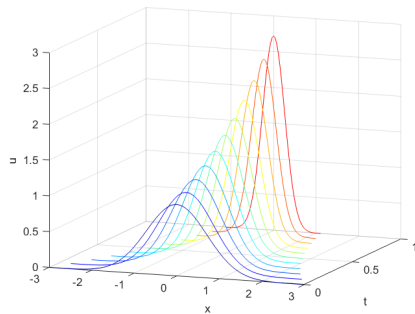
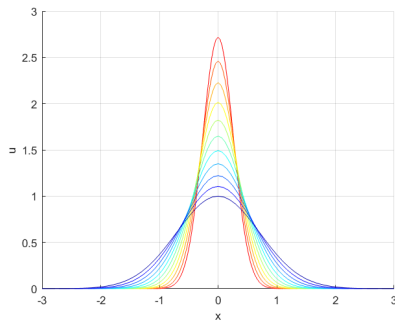
The **method of characteristics** is a technique for solving first-order partial differential equations (PDEs) by transforming them into a set of ordinary differential equations (ODEs) along specific curves called **characteristic curves**. Here's a straightforward approach to understanding this method.



Introduction

How is $u(x, t)$ changing from the perspective of the moving observer?

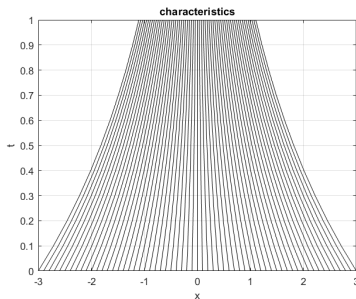
Let's define the position of a moving observer as $x(t)$, where the function $u(x, t)$ describes the system of interest. By examining the xu -plane, we can observe how u changes from the perspective of the observer, which is represented by the plot in the (x, u) coordinates.



Introduction

How do we see this curve changing?

If we fix x in the tx -plane, the resulting path $(x(t), t)$ corresponds to the characteristics of the system. Along these characteristics, we have access to all values of $u(x, t)$. To analyze how $u(x, t)$ changes as we move along this path, we utilize the first ordinary derivative, as the derivative inherently indicates the rate of change.



Introduction

Thus, we express this rate of change as follows:

$$\frac{d}{dt}u(x(t), t) = \frac{du}{dt} = \frac{\partial u}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial u}{\partial t} \cdot \frac{dt}{dt}$$

The next steps in deriving the method of characteristics depend on the type of first-order PDE we are working with. Different types of PDEs require slightly different approaches to derivation, and classifying the PDE helps guide the application of the characteristic method.

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The next steps in deriving the method of characteristics depend on the type of first-order PDE we are working with. Different types of PDEs require slightly different approaches to derivation, and classifying the PDE helps guide the application of the characteristic method.

Method of Characteristics for Semilinear PDEs: Consider a first-order semilinear partial differential equation (PDE) of the form

$$a(x, y)u_x + b(x, y)u_y = c(x, y, u),$$

where $a(x, y)$, $b(x, y)$, and $c(x, y, u)$ are given functions. We can reduce the problem to a system of ordinary differential equations (ODEs) as follows:

1. ****System of ODEs:**** Formally, we consider the system

$$\frac{dx}{a(x, y)} = \frac{dy}{b(x, y)} = \frac{du}{c(x, y, u)}.$$

2. ****Characteristic Curves:**** To find the characteristic curves, we solve the first pair of equations

$$\frac{dx}{a(x, y)} = \frac{dy}{b(x, y)} \Leftrightarrow \frac{dy}{dx} = \frac{b(x, y)}{a(x, y)} \Leftrightarrow \frac{dx}{dy} = \frac{a(x, y)}{b(x, y)}. \quad (3)$$

This will yield characteristic lines of the form $C = f(x, y)$.

3. ****General Solution:**** The general solution can then be obtained by solving either

$$\frac{dx}{a(x, y)} = \frac{du}{c(x, y, u)}$$

or

$$\frac{dy}{b(x, y)} = \frac{du}{c(x, y, u)}.$$

We select the ODE that is easier to solve. It may be necessary to use the characteristic curve and the implicit function theorem to express $f(x, y) = C$ as either $y = y(C, x)$ or $x = x(C, y)$ to eliminate a variable for solving the ODE.

4. ****Integration Constant:**** When integrating, the constant of integration should be expressed as a function

$$F(C) = F(f(x, y))$$

rather than as a standalone constant.

5. ****Initial Conditions:**** If initial conditions are provided, they can be used to determine the specific form of $F(C)$.

Example 1

Find the general solution to the given PDE:

$$u_t + 2u_x = 0$$

Solution:

1. Identify the functions for the given PDE:

$$a(t, x) = 1, \quad b(t, x) = 2, \quad c(t, x, u) = 0.$$

2. Formulate the system of ODEs:

$$\frac{dt}{1} = \frac{dx}{2} = \frac{du}{0}.$$

3. Solve each pair of ODEs:

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Example 1

3.1. First pair:

$$\frac{dt}{1} = \frac{dx}{2} \Rightarrow 2 dt = dx.$$

Integrate both sides:

$$\int 2 dt = \int dx \Rightarrow 2t + C_1 = x.$$

3.2. Second pair:

Since $\frac{du}{0} = 0$ implies that u is constant along characteristics, we conclude:

$$u = C_2.$$

Substitute $C_2 = F(C_1)$, where $C_1 = 2t - x$:

Thus, the general solution is of the form:

$$u = F(2t - x).$$

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Example 2

Find the general solution to the given PDE:

$$yu_y - xu_x = 2$$

Solution:

1. Identify the functions for the given PDE:

$$a(x, y) = -x, \quad b(x, y) = y, \quad c(x, y, u) = 2.$$

2. Formulate the system of ODEs:

$$\frac{dx}{-x} = \frac{dy}{y} = \frac{du}{2}.$$

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$$\frac{dx}{-x} = \frac{dy}{y} \Rightarrow y dx + x dy = 0 \Rightarrow C_1 = xy$$

Or, integrating both sides:

$$\begin{aligned} \int \frac{dx}{-x} &= \int \frac{dy}{y} \Rightarrow \ln C_1 - \ln x = \ln y \\ \Rightarrow \ln \frac{C_1}{x} &= \ln y \Rightarrow \frac{C_1}{x} = y \end{aligned}$$

3.2. Second pair:

On the other hand, from

$$\frac{dy}{y} = \frac{du}{2},$$

Integrating both sides, we obtain

$$\int \frac{dy}{y} = \int \frac{du}{2} \Rightarrow C_2 + 2 \ln y = u$$

Substitute $C_2 = F(C_1)$, where $C_1 = xy$:

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Geometric interpretation

Consider the following linear PDE with constant coefficients:

$$au_x + bu_y = 0$$

Here a and b are constants, such that $a^2 + b^2 \neq 0$, i.e., at least one of the coefficients is nonzero (otherwise, this would not be a differential equation).

Using the dot product in $\mathbb{R}^2$, we can rewrite the left-hand side of the PDE as

$$(a, b) \cdot (u_x, u_y) = 0, \quad \text{or equivalently,} \quad (a, b) \cdot \nabla u = 0.$$

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Geometric interpretation

This PDE can be interpreted in terms of the directional derivative of a function $u(x, y)$:

Denoting the vector $(a, b) = v$, we see that the left-hand side of the above equation is almost the directional derivative of u in the direction of the vector v , but it's not quite a unit vector. However, because the dot product equals zero, we can write:

$$D_v u(x, y) = \frac{1}{\sqrt{a^2 + b^2}}(a, b) \cdot \nabla u = 0.$$

Hence, the directional derivative in the direction of (a, b) is zero. This implies that the solution must be constant in the direction of v . (Why? Just consider that if the derivative of something is zero, then that quantity must be constant.)

This equation, $(a, b) \cdot \nabla u = 0$, implies that u must be constant along lines parallel to the vector (a, b) . So, what are the equations for these lines?

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This equation, $(a, b) \cdot \nabla u = 0$, implies that u must be constant along lines parallel to the vector (a, b) . So, what are the equations for these lines?

Geometric interpretation

To find them, we observe that lines parallel to (a, b) are given by

$$(x, y) \cdot (b, -a) = C,$$

where (x, y) lies on the line and $(b, -a)$ is perpendicular to (a, b) .

Since $(a, b) \cdot (b, -a) = ab - ab = 0$, we have

$$bx - ay = C,$$

which defines a **family of characteristic lines**, where C is a constant.

Thus, we conclude that the solution to the original problem is of the form:

$$u(x, y) = F(bx - ay),$$

where F is an arbitrary function.

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Characteristic Equation

Using the PDE to Derive the Characteristic Equations

Given a linear first-order PDE, such as

$$au_x + bu_y = 0,$$

we match the terms from the PDE to the terms in the total derivative from (3).
Setting

$$\frac{\partial u}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial u}{\partial t} \cdot \frac{dt}{dt} = 0,$$

with

$$\frac{dx}{dt} = a \quad \frac{dt}{dt} = b \quad \frac{du}{dt} = 0$$

we obtain

$$dt = \frac{dx}{a} = \frac{dt}{b} = \frac{du}{0}$$

The solution follows by simply solving two ODEs in the resulting system.

Example 3

Solve the IVP:

$$5u_x + 2u_y = 0$$

$$u(0, y) = \sin(y)$$

Solution:

1. Identify the functions for the given PDE:

$$a(x, y) = 5, \quad b(x, y) = 2, \quad c(x, y, u) = 0.$$

2. Formulate the system of ODEs:

$$\frac{dx}{5} = \frac{dy}{2} = \frac{du}{0}.$$

3. Solve each pair of ODEs:

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$$\frac{dx}{5} = \frac{dy}{2} \Rightarrow 2 dx = 5 dy.$$

Integrate both sides:

$$\int 2 dx = \int 5 dy \Rightarrow 2x = 5y + C_1.$$

Thus, $C_1 = 2x - 5y$.

3.2. Second pair:

Since $\frac{du}{0} = 0$ implies that u is constant along characteristics, we conclude:

$$u = C_2.$$

Substitute $C_2 = F(C_1)$, where $C_1 = 2x - 5y$:

Thus, the general solution is of the form:

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Finding F

From the initial condition $u(0, y) = \sin(y)$, we have:

$$u(0, y) = F(-5y) = \sin(y).$$

This implies that

$$F(Arg) = \sin\left(-\frac{Arg}{5}\right) = -\sin\left(\frac{Arg}{5}\right).$$

Expression for arbitrary function F

Since $Arg = 2x - 5y$, we substitute directly:

$$F(2x - 5y) = \sin\left(-\frac{2x - 5y}{5}\right) = -\sin\left(\frac{2x - 5y}{5}\right).$$

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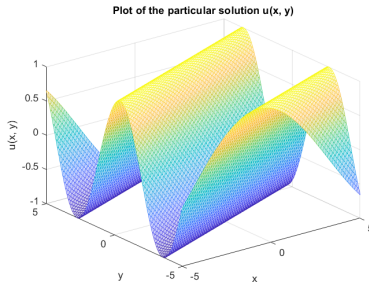
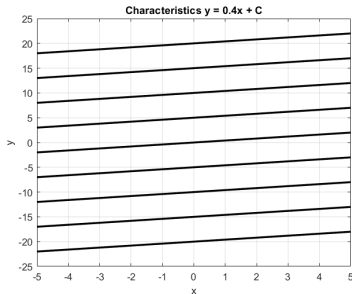
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Example 3

Final Solution

Hence, the correct final form should be:

$$u(x, y) = -\sin\left(\frac{2x - 5y}{5}\right).$$



Example 4

Solve the IVP:

$$2u_x + 3u_y = 0$$

$$u(x, 0) = x^2$$

Solution:

1. Identify the functions for the given PDE:

$$a(x, y) = 2, \quad b(x, y) = 3, \quad c(x, y, u) = 0.$$

2. Formulate the system of ODEs:

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From the initial condition $u(x, 0) = x^2$, we have:

$$u(x, 0) = F(3x) = x^2.$$

This implies that $x = \frac{Arg}{3}$.

$$F(Arg) = \left(\frac{Arg}{3}\right)^2 = \frac{Arg^2}{9}$$

Expression for arbitrary function F

Since $Arg = 3x - 2y$, we substitute directly:

$$F(3x - 2y) = \frac{(3x - 2y)^2}{9}.$$

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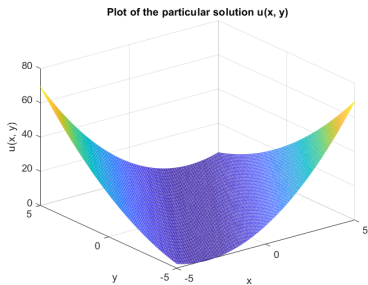
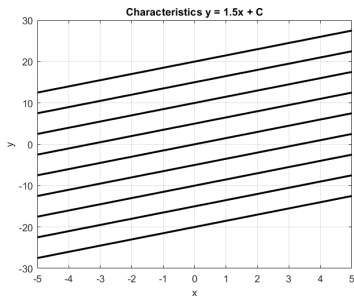
$$F(3x - 2y) = \frac{(3x - 2y)^2}{9}.$$

Example 4

Final Solution

Hence, the correct final form should be:

$$u(x, y) = \frac{(3x - 2y)^2}{9}.$$



Example 5

Solve the IVP:

$$u_t - 4u_x + u = 0$$

$$u(0, x) = \frac{1}{1 + x^2}$$

Solution:

1. Identify the functions for the given PDE:

$$a(x, t) = 1, \quad b(x, t) = -4, \quad c(x, t, u) = -u.$$

2. Formulate the system of ODEs:

$$\frac{dt}{1} = \frac{dx}{-4} = \frac{du}{-u}.$$

3. Solve each pair of ODEs:

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3.1. First pair:

$$\frac{dt}{1} = \frac{dx}{-4} \Rightarrow 4dt = -dx.$$

Integrate both sides:

$$\int 4dt = -\int dx \Rightarrow 4t = -x + C_1.$$

Thus, $C_1 = x + 4t$.

3.2. Second pair:

Since $\frac{du}{-u} = dt$ implies:

$$\ln |u| = -t + C_2 \Rightarrow u = C_2 e^{-t}.$$

Substitute $C_2 = F(C_1)$, where $C_1 = x + 4t$:

Thus, the general solution is of the form:

$$u = F(x + 4t)e^{-t}.$$

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Finding arbitrary function F

From the initial condition $u(0, x) = \frac{1}{1+x^2}$, we have:

$$u(0, x) = F(x)e^0 = F(x) = \frac{1}{1+x^2}.$$

Expression for F

Since $C_1 = x + 4t$, we substitute directly:

$$F(x + 4t) = \frac{1}{1 + (x + 4t)^2}.$$

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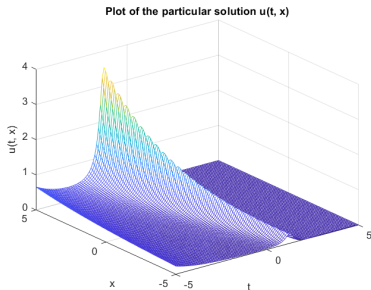
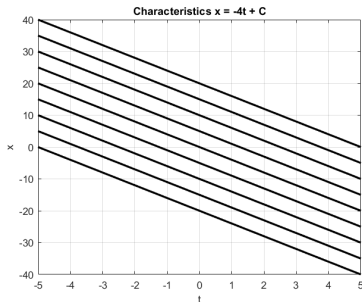
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Example 5

Final Solution

Hence, the final form should be:

$$u(t, x) = \frac{e^{-t}}{1 + (x + 4t)^2}.$$



Example 6

Solve the IVP:

$$\begin{aligned}xu_x + yu_y &= xe^{-u} \\ u(x, y) &= 0 \text{ on } y = x^2\end{aligned}$$

Solution:

1. Identify the functions for the given PDE:

$$a(x, y) = x, \quad b(x, y) = y, \quad c(x, y, u) = xe^{-u}.$$

2. Formulate the system of ODEs:

$$\frac{dx}{x} = \frac{dy}{y} = \frac{du}{xe^{-u}}.$$

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3.1. First pair:

This separable ODE leads to:

$$\frac{dy}{y} = \frac{dx}{x} \Rightarrow \ln |y| = \ln |x| + C_1 \Rightarrow y = C_1 x,$$

where C_1 is a constant.

3.2. Second pair:

For the equations $\frac{dx}{x} = \frac{du}{xe^{-u}}$, we have:

$$\int dx = \int e^u / du \Rightarrow e^u = x + C_2.$$

Substituting $C_1 = \frac{y}{x}$ into $C_2 = F(C_1)$:

Thus, the general solution is of the form:

$$u(x, y) = \ln \left(x + F \left(\frac{y}{x} \right) \right).$$

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From the initial condition $u(x, y) = 0$ when $y = x^2$, we have:

$$0 = u(x, y)|_{y=x^2} = \ln \left(x + F \left(\frac{x^2}{x} \right) \right) = \ln(x + F(x)).$$

This implies:

$$x + F(x) = 1 \quad \Rightarrow \quad F(x) = 1 - x.$$

Expression for $u(x, y)$

Substituting $F(x)$ back, we get:

$$u(x, y) = \ln \left(x + 1 - \frac{y}{x} \right).$$

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Quasilinear Equations: MoC

Consider the first-order quasilinear partial differential equation of the form

$$a(x, y, u)u_x + b(x, y, u)u_y = c(x, y, u), \quad (4.1)$$

subject to the initial condition

$$u(x, 0) = f(x), \quad x \in \mathbb{R}, \quad (4.2)$$

where a , b , and c are given real functions of the three variables x , y , and u . The function f is a given real function of the variable x , and u is the unknown function of two variables x and y .

A geometric interpretation of (4.1)

Assume that a smooth function $z = u(x, y)$ is a solution of (4.1). Then the graph $u = u(x, y)$ represents the solution surface S .

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Quasilinear Equations: MoC

The Normal Vector to the Surface S

Now, recall from vector calculus that the normal vector to the surface S is given by ∇f . Therefore, the equation for the normal vector to the surface $f(x, y, u) = 0$ is given by

$$\begin{aligned}\nabla f &= \frac{\partial f}{\partial x} \vec{i} + \frac{\partial f}{\partial y} \vec{j} + \frac{\partial f}{\partial u} \vec{k} = \\ &= \nabla f = \frac{\partial u(x, y)}{\partial x} \vec{i} + \frac{\partial u(x, y)}{\partial y} \vec{j} - \frac{\partial u(x, y)}{\partial u} \vec{k}\end{aligned}$$

or equivalently,

$$= \left(\frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}, -1 \right) = (u_x, u_y, -1)$$

Thus, the vector $\vec{n} = (u_x, u_y, -1)$ represents the **normal vector** at any point on the solution surface $u = u(x, y)$.

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Quasilinear Equations: MoC

Integral Surface

Further, we introduce the vector field

$$\mathbf{F} = a\vec{i} + b\vec{j} + c\vec{k},$$

where a , b , and c are the given coefficients of the quasilinear equation (4.1). Therefore, the PDE given by equation (4.1) can be written as $\mathbf{F} \cdot \nabla f = 0$, or

$$\begin{aligned}(a\vec{i} + b\vec{j} + c\vec{k}) \cdot \left(\frac{\partial u}{\partial x} \vec{i} + \frac{\partial u}{\partial y} \vec{j} - \vec{k} \right) &= 0, \\ &= (a, b, c) \cdot (u_x, u_y, -1) = 0\end{aligned}$$

This equation shows that the vectors (a, b, c) and $(u_x, u_y, -1)$ are **orthogonal**, and because $(u_x, u_y, -1)$ is normal to the surface $u = u(x, y)$, the vector $\mathbf{F} = (a, b, c)$ must be a **tangent vector** of the solution surface S at every point (x, y, u) .

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Integral Surface

Therefore, $\mathbf{F}$ defines a vector field in (x, y, u) space, that is tangent to the graphs of solutions of (4.1) at each point (x, y, u) .

Surfaces that are tangent to a vector field at each point in $\mathbb{R}^3$ are called **integral surfaces of the vector field**.

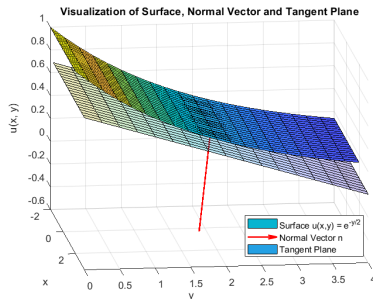
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Quasilinear Equations: MoC



To summarize, we have shown that $f(x, y, u) = u(x, y) - u = 0$, as a surface in the (x, y, u) space, is a solution of (4.1) if and only if the direction vector field $\mathbf{F} = (a, b, c)$ is contained in the tangent plane to the integral surface $f(x, y, u) = 0$ at each point (x, y, u) .

Quasilinear Equations: MoC

Since the vector $\mathbf{F}$ is tangent to the solution surface at every point (x, y, u) , we have the condition

$$\mathbf{F} \times d\mathbf{s} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ a & b & c \\ dx & dy & du \end{vmatrix} = 0$$

where $d\mathbf{s} = dx\vec{i} + dy\vec{j} + du\vec{k}$ is an elemental length along $\mathbf{F}$ on the solution surface.

Expanding the determinant, we obtain the following equation:

$$(b\,du - c\,dy)\hat{i} + (c\,dx - a\,du)\hat{j} + (a\,dy - b\,dx)\hat{k} = 0$$

which leads to (3):

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Quasilinear Equations: MoC

Characteristics

The equation

$$\frac{dx}{dy} = \frac{a(x, y, u)}{b(x, y, u)}$$

defines a family of curves (but dependent on u) in (x, y, u) space that lie on the solution surface. These curves are usually called **characteristics**; and the set of these equations is usually called the **characteristic equations** of the quasilinear PDE.

Projected Characteristic Curve

The projection of a characteristic curve onto the (x, y) plane ($u = 0$) is called a **characteristic base curve**, or **projected characteristic curve**, or simply **characteristic**.

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Example 1: Quasilinear Equations

Find the general solution to the given PDE:

$$2y \frac{\partial u}{\partial x} + u \frac{\partial u}{\partial y} = 2yu^2$$

Solution:

1. Identify the functions for the given PDE:

$$a = 2y, \quad b = u, \quad c = 2yu^2$$

2. Formulate the system of ODEs:

$$\frac{dx}{2y} = \frac{dy}{u} = \frac{du}{2yu^2}$$

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Example 1: Quasilinear Equations

3.1. Second pair:

$$\frac{du}{dy} = \frac{2yu^2}{u} = 2yu \quad (\text{for } u \neq 0)$$

Separating the variables and integrating to obtain

$$\int \frac{du}{u} = 2 \int y dy \Rightarrow \ln u = y^2 + \ln C_2$$

where C_2 is an arbitrary constant. The above equation can be rewritten as

$$u = C_2 e^{y^2}$$

3.2. First pair:

Plugging the value of u into the expression $\frac{dx}{dy} = \frac{dy}{u}$ yields

$$\frac{dx}{dy} = \frac{2y}{C_2 e^{y^2}}$$

which on integration gives

$$C_2 \int dx = \int 2y e^{-y^2} dy + C_1 \Rightarrow$$

$$C_2 x = -e^{-y^2} + C_1$$

where C_1 is the second arbitrary constant. Thus,

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$$C_2 \int dx = \int 2y e^{-y^2} dy + C_1 \Rightarrow$$

$$C_2 x = -e^{-y^2} + C_1$$

where C_1 is the second arbitrary constant. Thus,

$$C_2 x + e^{-y^2} = C_1, \quad u = C_2 e^{y^2}$$

Example 1: Quasilinear Equations

$C_2 = F(C_1)$:

Specifying C_2 as a function of C_1 defines a one-parameter family of characteristic curves, a solution surface. Thus, the general solution is expressed by writing $C_2 = F(C_1)$:, where

$$C_1 = C_2x + e^{-y^2} = ue^{-y^2}x + e^{-y^2} = (xu + 1)e^{-y^2}$$

which gives

$$u(x, y) = e^{y^2} f\left((1 + xu)e^{-y^2}\right)$$

an implicit relation for $u(x, y)$, where $F(\cdot)$ is an arbitrary function.

Remark:

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Parametric form of characteristic equations

It is often convenient to solve the characteristic equations and the associated Cauchy problem if various curves in the problem are represented in parametric form.

Let

$$x = x(s), \quad y = y(s), \quad u = u(s)$$

be the parametric equations of C in terms of the variable s .

Then the set of equations may be written as a system of ODE for $x(s)$, $y(s)$, and $u(s)$:

$$\frac{dx}{ds} = a \quad \frac{dy}{ds} = b \quad \frac{du}{ds} = c \tag{5}$$

where s actually gives a measure of the distance (arc length) along the curve.

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The system of ODE (10) is called *the characteristic equations of the PDE (4.1) in parametric form*. The family of curves $x = x(s)$, $y = y(s)$, $u = u(s)$ determined by solving the system of ODEs (5), are called the *characteristic curves* of the original PDE (4.1).

The existence and uniqueness theory of ODEs, assuming that certain smoothness conditions on the coefficients a , b , and c , guarantees that exactly one solution curve $(x(s), y(s), u(s))$ of (5) (i.e., a characteristic curve) passes through a given point (x_0, y_0, u_0) in (x, y, u) space.

Solution to IVP:

Often, we are not interested in the determination of a general solution of the PDE (4.1) but rather a specific solution $u = u(x, y)$ along a given curve (4.2) that passes through a given point (x, y) , typically (x_0, y_0) when $s = 0$.

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Consider the following types of the initial conditions and their parameterization.

Let the initial data (waveform) of u is given in the form:

$$u(x, 0) = F(x)$$

The value of function u at an arbitrary point x on the initial curve is then given by:

$$u(\xi, 0) = F(\xi).$$

The curve starts at the initial point $(x = \xi, y = 0)$, when $s = 0$, where the parameter s gives a measure of the distance along the curve.

Let the initial data (waveform) of u is given in the form:

$$u(0, y) = G(y)$$

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So, by solving the system of ODEs in parameterized form and utilizing the initial condition (also in parametric form), we obtain the solution of the initial value problem (IVP) (4.1-4.2) in the (ξ, s) plane:

$$x = X(s, \xi) \quad \text{and} \quad y = Y(s, \xi)$$

which describe the projected characteristics.

Projected Characteristics

However, it is often helpful to consider the projections of these characteristic curves onto the xy -plane. These **projected characteristics** are the curves traced out by $(x(s), y(s))$ as s varies, ignoring the z -component.

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Using these projections, we solve for s and ξ in terms of x and y , resulting in functions

$$s = S(x, y) \quad \text{and} \quad \xi = \Xi(x, y).$$

This allows us to locate where each point (x, y) lies along a specific characteristic in the xy -plane, as well as to obtain the solution to (4.1)-(4.2) in its explicit form.

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Example 2: Burgers's equation

Consider the initial-value problem for Burgers's equation

$$u_t + uu_x = 0, \quad u(x, 0) = 1 - x^2.$$

Step 1: Parametrization We parametrize our curve with s and ξ . The characteristic equations are:

$$\frac{dT}{ds} = 1, \quad \frac{dX}{ds} = Z, \quad \frac{dZ}{ds} = 0,$$

subject to the initial conditions $T(\xi, 0) = 0$, $X(\xi, 0) = \xi$, $Z(\xi, 0) = 1 - \xi^2$.

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Burgers' equation

Step 2: Solve the Characteristic Equations The solution of the above system is:

- From (1): $dT = ds$ implies $T(s) = s + C_1(\xi)$.
- From (3): $dZ = 0$ implies $Z(s) = C_3(\xi)$.
- From (2): $dX = Zds$ implies $dX = C_3(\xi)ds$ or $X(s) = C_3(\xi)s + C_2(\xi)$.

We can apply the initial conditions to find:

$$T(s) = s, \quad Z(s) = 1 - \xi^2, \quad X(s) = (1 - \xi^2)s + \xi.$$

Step 3: Solve for (s, ξ) in terms of (t, x) From the above, we have:

$$s = t, \quad x = (1 - \xi^2)t + \xi, \quad z = 0$$

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Step 4: Express $u(t, x)$ We rewrite u in terms of t and x :

$$u(t, x) = Z(s) = 1 - \xi^2.$$

From $x = (1 - \xi^2)t + \xi$, we find ξ as a function of t and x :

$$t\xi^2 - \xi + (x - t) = 0.$$

The solution for ξ is:

$$\xi = \frac{1 \pm \sqrt{1 - 4t(x - t)}}{2t}.$$

Thus, substituting ξ back into the expression for $u(t, x)$:

$$u(t, x) = 1 - \left(\frac{1 \pm \sqrt{1 - 4t(x - t)}}{2t} \right)^2 = \frac{2tx - 1 \pm \sqrt{1 + 4t^2 - 4tx}}{2t^2}$$

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